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Bus lanes make busy roads work better

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By Eric Murphy

In almost every urban area, congestion has been rising for years [despite expensive efforts to widen roads \(https://t4america.org/resource/congestion-con/\)](https://t4america.org/resource/congestion-con/) and add lanes for private vehicles. But [a new study \(https://doi.org/10.1016/j.tra.2026.104884\)](https://doi.org/10.1016/j.tra.2026.104884) shows that when demand is high, repurposing the road space already available can boost system performance without any expansion. Creating dedicated lanes for buses and bikes can optimize space – and failing to do so can start a vicious cycle of congestion.

When there's not much traffic, allocating more space to cars generally lets them move faster. But because they take up so much space, it doesn't take many more drivers to slow things down. Bikes and buses, however, can move more people without creating much traffic, making more efficient use of scarce space.

Without dedicated bus lanes, riders get the worst of both worlds: their bus will be stuck in the same traffic as if they were driving, leading to longer waits and longer trips. That encourages them to drive instead, making congestion worse. In contrast, setting aside dedicated road space for buses lets them move faster, encouraging drivers to take the bus instead. Researchers say at least 40% is the sweet spot on busy routes.

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“Allocating approximately half the road space to buses is critical,” says the new study. The authors explain that when buses with less dedicated space move slower, fewer people take them. When space is reallocated based on demand, less space for buses means slower rides and even fewer riders, in a vicious cycle. “Conversely,” say the authors, “if buses are allocated more space initially, more passengers are attracted to the bus due to shorter waiting times, generating a virtuous cycle.”

As we’ve previously written, [bus lanes also make the road safer](https://ssti.us/2024/08/05/dedicated-bus-lanes-improve-safety/) (<https://ssti.us/2024/08/05/dedicated-bus-lanes-improve-safety/>), even when buses aren’t present. The busy arterial corridors where bus lanes can be most effective are also where many cities have their biggest traffic safety issues.

The study also found that when there is moderate traffic, creating more space for bikes can improve road performance. They say planners should be proactive when deciding how much space to give each potential road user, rather than reacting to how people currently use the system – “[designing for the city they want](https://ssti.us/2025/12/08/we-dont-need-to-overthink-induced-demand-to-act-on-it/) (<https://ssti.us/2025/12/08/we-dont-need-to-overthink-induced-demand-to-act-on-it/>) instead of the city they have.”

Finally, the researchers say that in addition to dedicated bus lanes, allowing buses “limited, well-managed access to general-purpose lanes” during periods of high demand can further reduce travel times and complement the benefits of bus-only lanes to system performance.

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